

# Project XTC

## X-Touch Companion

The Unofficial Guide to the Behringer X-Touch, Bitwig Studio and DrivenByMoss

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# Reference

## Quick Reference

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## Chapter 1 - Meet the X-Touch

### Why "Project XTC"?

XTC stands for X-Touch Companion. This guide is designed to sit alongside the official Behringer, Bitwig and DrivenByMoss documentation, bringing everything together into one practical handbook. If the name raises a knowing smile, that's perfectly fine - but throughout this book, XTC always means X-Touch Companion.

If you've bought a Behringer X-Touch, you've probably already discovered two things. First, it feels like a serious piece of studio equipment. Second, learning to use it effectively with Bitwig can be surprisingly difficult.

The Behringer manual explains the hardware. Bitwig explains the DAW. DrivenByMoss explains the extension. What has been missing is a guide that explains how all three work together. That is the purpose of Project XTC.

### Who This Book Is For

This guide assumes you already know the basics of using a computer and Bitwig Studio. It does not assume any previous knowledge of MCU, MIDI protocols or control surfaces. Every important idea is introduced only when it becomes useful.

### What the X-Touch Is

The X-Touch is a hardware control surface. It does not process audio or replace Bitwig. It provides tactile control over track selection, transport, mixing, PLUG-IN, automation and much more.

### The Four Principles

1. Bitwig is in charge.
2. The selected track matters.
3. Modes change meanings.
4. Feedback is as important as input.

## Before You Continue...

Don't worry about memorising every button yet. In the next chapter we'll take a guided tour of the X-Touch so every control has a name and a purpose before we start using it.

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## Chapter 2 - A Tour of the Hardware

Before learning individual features, it's worth becoming familiar with the physical layout of the X-Touch. You don't need to memorise every control. The goal is simply to help you recognise where things are, so later chapters can refer to them naturally.

### The Eight Channel Strips

The heart of the X-Touch consists of eight identical channel strips. Each strip includes a motor fader, a touch-sensitive V-Pot with an LED ring, a scribble strip display, and buttons such as SELECT, SOLO, MUTE and REC. Because every strip is identical, once you understand one, you understand them all.

### The Master Section

To the right of the eight channels is the master section. It includes the master fader and additional navigation controls.

### Assignment Buttons

The TRACK, SEND, PAN, PLUG-IN, EQ and INSTRUMENT buttons change what the V-Pots control.

### GLOBAL VIEW

The USER button in GLOBAL VIEW opens Browser Mode when using DrivenByMoss.

### Reality Check

If the number of buttons feels intimidating, remember that you rarely use them all at once.

### Before You Continue...

The next chapter introduces the mental model that makes the X-Touch predictable.

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## Chapter 3 - The Mental Model

Most difficulties people experience with the X-Touch come from thinking of it as a mixer with fixed controls. It isn't. The X-Touch is better understood as a window into Bitwig.

### Bitwig Is Always in Charge

The X-Touch follows Bitwig's current state. If Bitwig changes, the controller reflects that change.

### Selection Comes First

Before adjusting anything ask: What is currently selected? Successful workflows nearly always begin with SELECT.

### Modes Change Meaning

The same controls perform different jobs in different contexts. Understanding modes is more important than memorising combinations.

### Always Read the Feedback

Motor faders, LED rings and scribble strips constantly tell you what Bitwig is doing.

### A Simple Workflow

SELECT -> Observe -> Adjust -> Listen

Notice that adjustment comes third, not first.

#### Reality Check

If the X-Touch behaves unexpectedly, check the selected track, current mode and scribble strips first.

#### Field Note

The best X-Touch users don't memorise hundreds of functions. They ask, "What is Bitwig telling me right now?"

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## Chapter 4 - Banks and Channels

One of the first things to understand about the X-Touch is that it always provides eight physical channel strips, regardless of how many tracks exist in your Bitwig project.

If your project contains more than eight tracks, the X-Touch simply changes which eight tracks are currently under your control. This is where the ideas of channels and banks become important.

### Channels

Each motor fader represents a channel.

In the simplest case:

| X-Touch   | Bitwig  |
|-----------|---------|
| Channel 1 | Track 1 |
| Channel 2 | Track 2 |
| Channel 3 | Track 3 |
| ...       | ...     |
| Channel 8 | Track 8 |

If your project contains eight tracks or fewer, this relationship never changes.

The X-Touch behaves very much like a traditional hardware mixer.

### Banks

Once your project grows beyond eight tracks, the controller cannot display every track at once.

Instead, it displays a bank of eight consecutive tracks.

The first bank contains:

- Tracks 1-8



The second bank contains:

- Tracks 9–16

The third bank contains:

- Tracks 17–24

and so on.

Changing bank simply changes which group of eight tracks is currently mapped onto the eight physical channel strips.

Field Note

The hardware never changes.

Only the relationship between the hardware and the Bitwig project changes.

## BANK Buttons

The BANK ◀ and BANK ▶ buttons move one complete bank at a time.

For example, if you are currently controlling Tracks 1–8:

- Press BANK ▶ once.
- The scribble strips now display Tracks 9–16.
- The motor faders immediately move to the stored levels for those tracks.

Press BANK ◀ to return to the previous bank.

Watching the motor faders reposition themselves is one of the most satisfying demonstrations of what a motorised control surface can do.

## CHANNEL Buttons

The CHANNEL ◀ and CHANNEL ▶ buttons work differently.

Instead of moving by eight tracks, they move by one track.

Imagine you are currently controlling:

| Fader | Track   |
|-------|---------|
| 1     | Track 1 |
| 2     | Track 2 |
| 3     | Track 3 |
| 4     | Track 4 |

| Fader | Track   |
|-------|---------|
| 5     | Track 5 |
| 6     | Track 6 |
| 7     | Track 7 |
| 8     | Track 8 |

Press CHANNEL ► once.

Now the mapping becomes:

| Fader | Track   |
|-------|---------|
| 1     | Track 2 |
| 2     | Track 3 |
| 3     | Track 4 |
| 4     | Track 5 |
| 5     | Track 6 |
| 6     | Track 7 |
| 7     | Track 8 |
| 8     | Track 9 |

Every channel shifts along by one track.

This makes it easy to bring a particular track into view without jumping an entire bank.

#### Reality Check

If a track seems to have "disappeared", it has probably moved outside the currently visible bank.

Use the BANK or CHANNEL buttons to bring it back into view.

## Reading the Scribble Strips

Whenever you change bank or channel offset, the scribble strips update immediately.

Experienced users naturally glance at the displays before touching a fader.

This simple habit avoids accidental edits to the wrong track.

## Why This Matters

Understanding banks and channels makes the rest of the X-Touch much easier to understand.

Whether you are controlling volume, sends, plug-ins or automation, the same principle always applies:

The X-Touch controls the tracks that are currently visible.

Everything else in the controller builds upon this idea.

### Exercise

Create a Bitwig project containing at least twelve tracks.

Use the BANK ► and BANK ◀ buttons to move between the first and second banks.

Then experiment with the CHANNEL ► button and watch how the scribble strips and motor faders update.

Pay particular attention to which track appears on Channel 1 after each movement.

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## Chapter 5 - Modes

The X-Touch is often described as having "lots of buttons". That is true, but it is only half the story.

The more important fact is that many of those buttons do different jobs at different times.

Understanding modes is the key to understanding the X-Touch.

Once this idea becomes familiar, the controller feels logical rather than mysterious.

### What Is a Mode?

A mode changes the meaning of one or more controls.

The physical hardware never changes.

Instead, the X-Touch changes how it interprets your button presses, encoder turns and fader movements.

Think of the controller as speaking several different "languages".

Each mode uses the same hardware to perform a different set of tasks.

#### Field Note

A common mistake is to think that every button has exactly one purpose.

On the X-Touch, many controls are intentionally multifunctional.

Figure 5.1 illustrates this idea. Although the hardware remains exactly the same, selecting a different mode changes the role of the controls rather than the controls themselves.

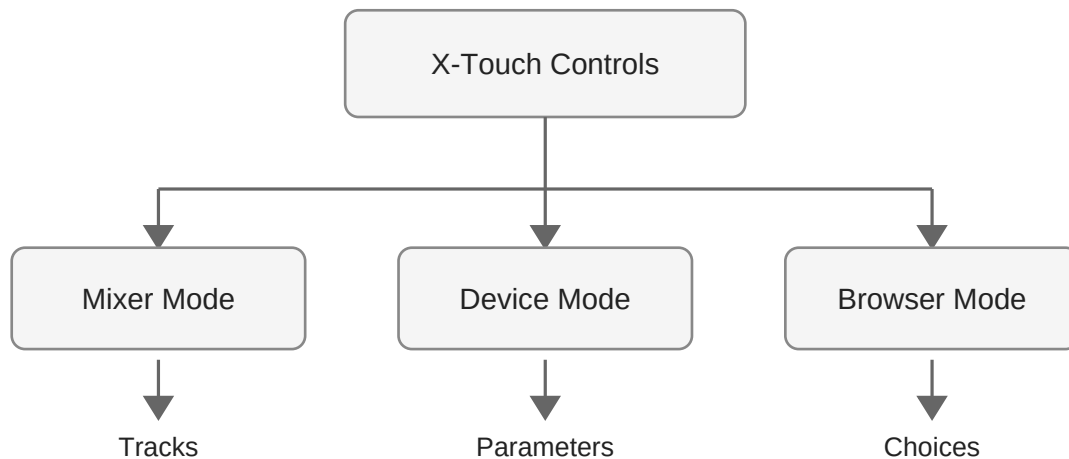


Figure 5.1 — The X-Touch uses the same physical controls in many different operating modes.

## The Current Mode Is Always Visible

Fortunately, the X-Touch rarely leaves you guessing.

The scribble strips, LEDs and button illumination usually indicate the current operating mode.

Before pressing a control, take a quick look at the feedback the controller is already providing.

Experienced users develop this habit very quickly.

## The Primary Modes

DrivenByMoss makes extensive use of the X-Touch's mode system.

The most frequently used modes include:

- TRACK
- SEND
- PAN/SURROUND
- PLUG-IN
- EQ
- INSTRUMENT
- USER

Each mode changes what the rotary encoders control and, in some cases, what appears on the scribble strips.

The important point is that these are not different layouts of the controller. They are simply different interpretations of the same hardware.

Throughout the remainder of this book we shall examine each of these modes in detail.

## TRACK Mode

TRACK mode is the mode most users spend the majority of their time in.

The controller focuses on the Bitwig tracks themselves.

Typical operations include:

- selecting tracks
- adjusting volume
- muting
- soloing
- recording
- automation

Think of TRACK mode as the controller's "home".

Whenever you become unsure where you are, returning to TRACK mode is often a sensible starting point.

## SEND Mode

SEND mode changes the rotary encoders so that they adjust send levels instead of pan.

Instead of controlling the stereo position of each track, the encoders now control how much of the signal is sent to an effects bus such as a reverb or delay.

Exactly which send is being adjusted depends upon the current Bitwig project and the selected send.

### Reality Check

If turning an encoder suddenly changes a reverb or delay send instead of the pan position, the controller is almost certainly in SEND mode.

## PAN/SURROUND Mode

PAN/SURROUND mode returns the rotary encoders to controlling the stereo (or surround) position of each track.

For most stereo projects this simply means moving sounds left or right within the mix.

## PLUG-IN Mode

PLUG-IN mode is one of the X-Touch's most powerful capabilities.

Rather than controlling the mixer, the encoders are assigned to parameters exposed by the currently selected Bitwig device.

The scribble strips display parameter names, allowing the hardware to become a tactile extension of the software interface.

Later chapters explore this workflow in depth.

## INSTRUMENT Mode

INSTRUMENT mode provides direct access to software instrument parameters.

Depending upon the selected Bitwig device, the encoders may control filter cutoff, resonance, envelopes, oscillator settings or many other synthesis controls.

The exact assignments depend entirely upon the instrument currently in focus.

## USER Mode

USER mode deserves special attention.

Unlike the other operating modes, USER mode is not tied to a particular mixer function.

DrivenByMoss uses USER mode to expose additional Bitwig features.

For example, pressing USER immediately opens Bitwig's Browser. The scribble strips are repurposed to display browser information, allowing sounds and presets to be selected directly from the X-Touch.

### Field Note

Because USER mode is completely software-defined, it is capable of doing far more than its name might suggest.

Many users overlook it at first, yet it quickly becomes one of the most frequently used buttons on the controller.

## Modes Are Temporary

Changing mode does not permanently alter the controller.

It simply changes what the controls do at that moment.

You are free to move between modes whenever your workflow requires.

The motor faders, scribble strips and LEDs update automatically to reflect the current context.

## Thinking in Modes

One of the biggest steps towards mastering the X-Touch is learning to ask a simple question:

"What mode am I in?"

Many apparent "problems" disappear the moment you realise that the controller is behaving exactly as expected—it is simply operating in a different mode.

Once this becomes second nature, the X-Touch feels far less complicated than it first appears.

### Exercise

Open any Bitwig project.

Press each of the available mode buttons in turn and observe how the scribble strips, encoder assignments and button illumination change.

Do not try to memorise every function immediately.

Instead, become comfortable with the idea that the X-Touch continually adapts its controls to the task at hand.

Recognising the current mode is far more valuable than memorising individual button assignments.



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## Chapter 6 - The SELECT Button

The SELECT button is arguably the most important button on the X-Touch.

New users often assume that its only purpose is to highlight a track. In fact, SELECT determines where the controller's attention is focused. Many of the X-Touch's most powerful features depend upon this simple action.

Once you understand SELECT, the behaviour of Device Mode, Browser Mode and many other functions becomes much easier to understand.

### Selecting a Track

Each channel strip has its own SELECT button.

Pressing one tells the X-Touch:

"This is the track I want to work with."

Nothing is changed simply by selecting a track.

Instead, you are telling the controller where future actions should take place.

Think of SELECT as moving your attention rather than changing your project.

#### Field Note

The X-Touch always has a current focus.

Many commands operate on the currently selected track, so selecting the correct track before performing another action quickly becomes second nature.

### Focus Rather Than Selection

Although the button is labelled SELECT, it is often more helpful to think of it as a FOCUS button.

When you select a different track:

- Device Mode follows that track.
- Browser Mode follows that track.

- Plug-in parameters follow that track.
- Instrument controls follow that track.

The selected track becomes the centre of attention for the controller.

This idea of focus appears repeatedly throughout the rest of the book.

## Watching the Controller Respond

One of the strengths of DrivenByMoss is the excellent feedback it provides.

After pressing a SELECT button you may notice changes to:

- the scribble strips
- encoder assignments
- parameter names
- button illumination
- device information

These are not separate operations.

They are all consequences of changing the controller's focus.

The X-Touch immediately adapts to the newly selected track.

## SELECT Comes First

As you become more familiar with the controller, a natural workflow begins to emerge:

1. Select the track.
2. Choose the required mode.
3. Perform the adjustment.

Whether you are controlling a plug-in, adjusting sends or browsing presets, the first step is almost always the same.

Press SELECT.

### Reality Check

If the X-Touch appears to be controlling the "wrong" plug-in or instrument, the most likely explanation is that a different track is currently selected.

Before assuming something is broken, check which SELECT button is illuminated.

## SELECT and the Motor Faders

A common misunderstanding is that selecting a track should cause the motor faders to move.

They do not.

The motor faders represent the currently visible bank of channels.

SELECT changes the controller's focus, not the mixer layout.

This distinction is important.

The faders show where the tracks are.

SELECT shows which one you are working on.

## SELECT in Device Mode

Device Mode always follows the selected track.

Imagine a project containing:

- drums
- bass
- piano
- synthesizer
- vocals

Each track contains different devices.

Simply pressing SELECT on another channel instantly changes which devices the controller is connected to.

The rotary encoders, parameter names and scribble strips all update automatically.

No additional configuration is required.

## SELECT in Browser Mode

Browser Mode behaves in exactly the same way.

The currently selected track determines where new devices, presets or samples will be inserted.

This makes SELECT an important part of everyday workflow.

Rather than opening the Browser and then deciding where something should go, experienced users select the destination track first.

## Developing Good Habits

The most effective X-Touch users develop a very simple habit:

SELECT → Observe → Adjust

Notice that adjustment comes last.

First establish the correct focus.

Then observe what the controller is telling you.

Only then make changes.

This small change in thinking prevents many mistakes.

### Field Note

One of the recurring themes throughout this book is that the X-Touch is constantly providing feedback.

The more you learn to observe that feedback before pressing buttons, the more natural the controller becomes.

## A Typical Workflow

A common sequence might look like this:

1. Press SELECT on the desired track.
2. Press PLUG-IN.
3. Adjust the required parameters.
4. Press TRACK to return to mixer control.

Although simple, this workflow is repeated countless times during a typical production session.

It quickly becomes second nature.

## The Bigger Picture

At first glance, SELECT appears to be a very ordinary button.

In reality, it sits at the centre of the X-Touch's operating philosophy.

The controller is always asking the same question:

"Which track am I working on?"

The SELECT button provides the answer.

Understanding this idea of focus is one of the biggest steps towards mastering the X-Touch.

## Exercise

Open a Bitwig project containing several tracks.

Press the SELECT button on each channel strip in turn.

After each selection:

- enter Device Mode
- enter Browser Mode
- observe the scribble strips
- observe the encoder assignments

Notice that the hardware itself never changes.

Only the controller's focus changes.

Once you begin thinking in terms of focus rather than simply selection, many other features of the X-Touch become much easier to understand.

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## Chapter 7 - Displays and Feedback

The X-Touch is much more than a collection of switches, knobs and motor faders.

It is constantly communicating with you.

Almost every action performed in Bitwig is reflected somewhere on the controller. Likewise, almost every action you perform on the controller is reflected in Bitwig.

Understanding this two-way conversation is one of the keys to using the X-Touch confidently.

### More Than a Remote Control

Many MIDI controllers simply send commands to the computer.

The X-Touch goes much further.

It continuously receives information from Bitwig and updates its displays, LEDs and motor faders to match the current state of the project.

Rather than being a passive controller, it becomes an extension of the software itself.

#### Field Note

A useful way to think about the X-Touch is that it is a window into Bitwig.

Instead of constantly looking at the computer monitor, you can often see exactly what is happening simply by glancing at the controller.

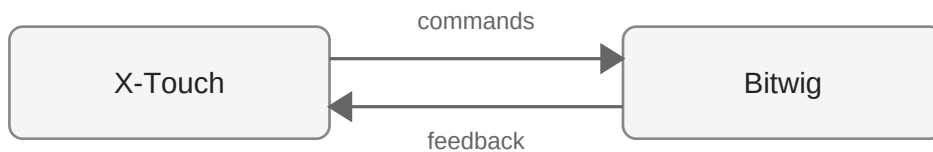


Figure 7.1 — Information flows continuously in both directions between Bitwig and the X-Touch.

## Scribble Strips

The eight LCD scribble strips are one of the X-Touch's defining features.

Most of the time they display track names, but that is only one of their many roles.

Depending upon the current operating mode they may display:

- track names
- device names
- plug-in parameters
- browser entries
- send names
- automation information

The important point is not to memorise every display.

Instead, remember that the scribble strips are always showing the information that is most relevant to your current task.

## LED Rings

Surrounding each rotary encoder is an LED ring.

These provide immediate visual feedback about the value being controlled.

Different parameters use different display styles.

For example:

- pan controls are centred
- continuous values fill progressively
- switches simply indicate on or off

After a short time you begin to recognise these patterns instinctively.

Often you can tell what a control is doing before you even touch it.

## Illuminated Buttons

Many buttons on the X-Touch include built-in LEDs.

These indicate the current state of important functions such as:

- MUTE
- SOLO
- RECORD
- transport operations
- automation modes

The illuminated buttons are constantly reflecting the current state of Bitwig.

They are not merely indicators of what you have pressed.

## Motor Faders

The motor faders provide perhaps the most dramatic form of feedback.

Whenever you load a project, switch banks or recall automation, the faders immediately move to their new positions.

This movement is not simply a visual effect.

It confirms that the controller has received updated information from Bitwig and is synchronised with the project.

### Reality Check

Never try to prevent a motor fader from moving.

The motors are positioning the fader to match Bitwig's current value.

Allow them to move freely.

## Feedback Builds Confidence

Imagine selecting a different track.

Within a fraction of a second you may notice:

- new track names
- different plug-in parameters
- updated LED rings
- illuminated buttons changing state



All of this happens automatically.

The controller is telling you that its focus has changed.

The more you learn to trust this feedback, the more natural the X-Touch becomes.

## Observe Before You Adjust

One habit separates experienced users from beginners.

Beginners often press buttons first and try to understand the result afterwards.

Experienced users do the opposite.

Before making any adjustment they glance at the controller and ask:

- Which track is selected?
- Which mode am I in?
- What are the scribble strips showing?
- What do the LED rings indicate?
- Which buttons are illuminated?

Only then do they begin making changes.

This simple habit avoids countless mistakes.

### Field Note

Throughout this book we return to a single idea:

Observe before you adjust.

The X-Touch is continually telling you what it is doing.

Learning to read that information is more valuable than memorising button combinations.

## The Controller Is Always Working

Even when you are not touching it, the X-Touch is active.

Automation moves.

Track selection changes.

Devices change.

Projects load.

Throughout all of this the controller quietly updates itself to remain synchronised with Bitwig.

That constant feedback is one of the reasons the X-Touch feels so different from simpler MIDI controllers.

## Looking Ahead

The next chapters explore the controller's physical controls in more detail.

As you learn about the rotary encoders, motor faders and transport controls, remember that every control performs two roles:

It accepts your input.

It also provides valuable feedback.

Understanding both roles is essential to mastering the X-Touch.

### Exercise

Open any Bitwig project and spend two minutes simply observing the controller.

Without pressing any buttons, identify:

- the selected track
- the current mode
- the information shown on the scribble strips
- the LED ring positions
- illuminated buttons
- motor fader positions

Now select a different track.

Watch how the controller updates before touching anything else.

Developing the habit of reading the controller before operating it will make every subsequent chapter easier to understand.

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## Chapter 8 - Modifiers: SHIFT, OPTION, CONTROL and ALT

### Modifiers: SHIFT, OPTION, CONTROL and ALT

So far, most of the controls in this guide have been considered one at a time.

Press a button.

Turn a V-Pot.

Move a fader.

But the X-Touch has far more functions than it has physical controls.

One of the ways DrivenByMoss solves this is through modifier buttons.

The four important modifiers are:

- SHIFT
- OPTION
- CONTROL
- ALT

Hold one of these while operating another control, and the second control may perform a different function.

If you are familiar with keyboard shortcuts, the principle is already familiar:

control  
+  
modifier  
=  
another function

The important thing is not to memorise every possible combination immediately.

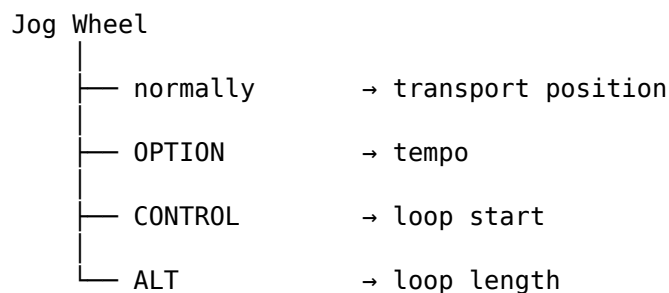
It is to understand the pattern.

## One Surface, More Functions

Consider a control we already know: the Jog Wheel.

Turn it normally and it changes the transport position.

But DrivenByMoss can reinterpret that same physical movement when a modifier is held:



The Jog Wheel has not physically changed.

Its context has.

This is the same principle introduced earlier in this guide when we looked at modes.

A modifier creates a temporary context.

## Modifiers Are Temporary Modes

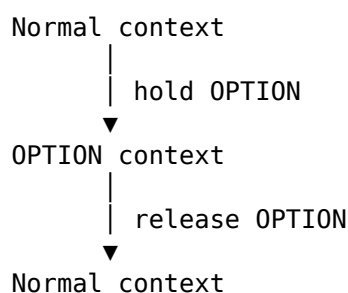
There is a useful way to think about the four modifier buttons:

Holding a modifier temporarily changes the meaning of another control.

A normal mode may remain active until you leave it.

A modifier usually lasts only while you hold its button.

Conceptually:



This makes modifiers particularly useful for secondary operations.

The main function remains immediately available, while the less frequently needed function sits behind a modifier.

## SHIFT

SHIFT is probably the modifier you will encounter most often.

It frequently provides:

- a secondary version of an operation;
- finer control;
- reverse movement through a sequence.

For example, when adjusting some continuous parameters, SHIFT gives finer resolution.

With the Jog Wheel:

Jog Wheel  
→ move transport position

SHIFT + Jog Wheel  
→ move transport position more finely

In other contexts SHIFT selects a related secondary operation.

You will see examples throughout the following chapters.

Do not assume that SHIFT always means "fine adjustment", however.

Its exact meaning depends on the control and the current mode.

## OPTION

OPTION often changes what the operation acts upon or exposes a related operation.

For example, BANK and CHANNEL normally move the controller's view through the project.

With OPTION held, DrivenByMoss can instead use those controls to move objects within the project.

Conceptually:

```

BANK / CHANNEL
├── normally
│   move the VIEW
└── OPTION
    move an OBJECT
  
```

OPTION is also important in several workflows we will meet later.

For example:

OPTION + MARKER

creates a marker.

And:

OPTION + REWIND

OPTION + FORWARD

navigate between markers.

OPTION therefore does not have one universal definition.

Think of it as:

Give this control its alternative operation.

## CONTROL

CONTROL frequently exposes a structural operation or another dimension of control.

One particularly useful example appears in Device Mode.

Holding CONTROL allows the V-Pots to expose devices so that a particular device can be selected directly.

CONTROL also changes the Jog Wheel:

CONTROL + Jog Wheel

adjusts the loop start.

With SHIFT added:

CONTROL + SHIFT + Jog Wheel

provides finer adjustment.

Again, CONTROL does not have one fixed meaning.

The current mode still matters.

## ALT

ALT is another modifier used to expose alternative parameter operations.

With the Jog Wheel:

ALT + Jog Wheel

adjusts loop length.

Adding SHIFT gives finer control:

ALT + SHIFT + Jog Wheel

adjusts loop length more precisely.

ALT also appears in other specialised workflows, including operations concerned with clip length.

These will make more sense when we reach the relevant chapters.

## Modifiers Can Be Combined

Modifiers are not necessarily used alone.

SHIFT can be combined with another modifier.

The Jog Wheel gives us an excellent example:

| Control                     | Function                |
|-----------------------------|-------------------------|
| Jog Wheel                   | Transport position      |
| SHIFT + Jog Wheel           | Fine transport position |
| OPTION + Jog Wheel          | Tempo                   |
| OPTION + SHIFT + Jog Wheel  | Fine tempo              |
| CONTROL + Jog Wheel         | Loop start              |
| CONTROL + SHIFT + Jog Wheel | Fine loop start         |
| ALT + Jog Wheel             | Loop length             |
| ALT + SHIFT + Jog Wheel     | Fine loop length        |

There is a pattern here.

SHIFT does not change what OPTION, CONTROL or ALT selects.

Instead, it changes the precision with which that parameter is adjusted.

So:

OPTION

↓

Tempo

OPTION + SHIFT

↓

Tempo, but finer

This is exactly the sort of relationship worth learning.

It reduces the number of apparently unrelated commands you need to remember.

## Modifier Patterns, Not Modifier Rules

You may already have noticed some tendencies:

SHIFT

→ secondary / finer / reverse

OPTION

→ alternative operation

CONTROL

→ structural or additional control

ALT

→ another specialised parameter

These are useful mental shortcuts.

But they are not rules.

DrivenByMoss assigns modifier combinations according to what is useful in a particular context.

For example:

OPTION + MARKER

creates a marker.

There is no useful way to derive "create marker" from a universal definition of OPTION.

You simply learn that command as part of the Marker workflow.

The purpose of recognising modifier patterns is therefore not to predict every command.

It is to make the controller easier to understand when you encounter one.

## Order Matters Less Than the Combination

When this guide writes:

OPTION + MARKER

it means:

1. hold OPTION;
2. press MARKER;
3. release the buttons.

Likewise:

CONTROL + SHIFT + Jog Wheel

means:

1. hold CONTROL;
2. hold SHIFT;
3. turn the Jog Wheel;
4. release the modifiers.

The `+` notation means use these controls together.



It does not mean that you should press them as a rapid sequence.

## Modifier Buttons and Modes Work Together

Modifiers do not replace modes.

They work inside them.

This distinction is important.

Suppose the controller is in Device Mode.

The V-Pots already have meanings determined by Device Mode.

Now hold OPTION.

The V-Pots can acquire another set of meanings appropriate to that mode.

So the complete context may be thought of as:

```
Physical control
      +
Current mode
      +
Held modifier
      =
Current function
```

This extends the mental model from Chapter 3.

The function of a physical control is not necessarily inherent in the control itself.

It emerges from context.

## The Display Is Your Ally

Modifier combinations can make the surface seem complicated if you try to remember everything from button labels alone.

Don't.

As discussed in Chapter 7, the displays are part of the control system.

When a modifier or mode changes what controls mean, look at the feedback the X-Touch and Bitwig provide.

The controller should be treated as a conversation:

```
You change context
      ↓
DrivenByMoss changes assignments
      ↓
The displays provide feedback
      ↓
You make the next decision
```

This becomes especially important in Device Mode, Browser Mode and the advanced edit modes.

## Don't Learn the Modifier Table

It is tempting at this point to make a huge table containing every possible combination of:

SHIFT  
OPTION  
CONTROL  
ALT

with every button on the X-Touch.

That would turn this guide into exactly the kind of manual we are trying to avoid.

Instead, learn modifier combinations with the task they perform.

When we study markers, we will learn:

OPTION + MARKER

because it is part of the marker workflow.

When we study Device Mode, we will learn CONTROL and OPTION combinations because they help navigate devices and parameter pages.

When we study recording, we will learn the RECORD modifiers because they make sense in the context of recording.

The modifier is not the subject.

The job you are trying to do is the subject.

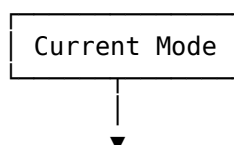
## A Useful Mental Model

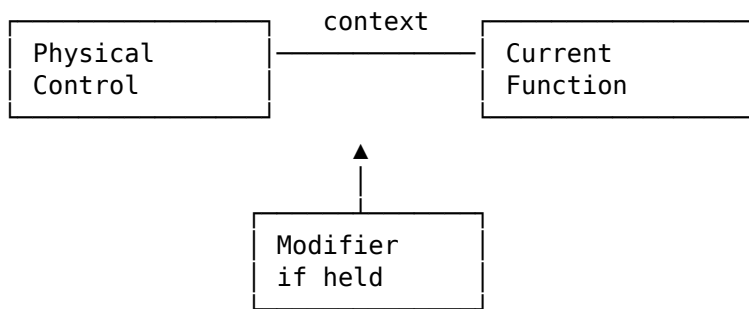
At this point we can expand our model of the X-Touch considerably.

Earlier we had:

physical control  
↓  
current mode  
↓  
current function

Now we can add modifiers:





You do not need to consciously think through this diagram every time you touch the controller.

Its purpose is to explain why the same V-Pot, button or wheel can perform so many different jobs without the surface becoming physically enormous.

## What Comes Next

Now that we understand modifiers, the next chapters can use them without stopping repeatedly to explain what SHIFT, OPTION, CONTROL and ALT are.

We will see them applied to:

- V-Pots;
- motor faders;
- transport;
- the Jog Wheel;
- devices;
- Browser navigation;
- mixer modes;
- markers;
- automation;
- groups;
- recording.

The individual combinations will be introduced when they become useful.

## The Important Idea

If there is one thing to take away from this chapter, it is this:

A modifier temporarily changes the context of another control.

You do not need to memorise every modifier combination.

Learn the basic controls first.

Then learn the modified versions as part of real workflows.

Once that pattern becomes familiar, SHIFT, OPTION, CONTROL and ALT stop looking like four more sets of commands to memorise.

They become something much more useful:

four ways of asking the X-Touch to do a little more.

# Project XTC

## X-Touch Companion

The Unofficial Guide to the Behringer X-Touch, Bitwig Studio and DrivenByMoss

## Chapter 9 - V-Pots (Rotary Encoders)

The eight rotary encoders above the channel strips are among the most versatile controls on the X-Touch.

Behringer refers to them as V-Pots.

At first glance they look like ordinary rotary knobs. In reality, each V-Pot combines three useful features:

- a rotary encoder
- a push switch
- an LED ring

More importantly, their purpose changes according to the current mode.

A V-Pot might control pan in one moment, a send level in another, and a plug-in parameter a few seconds later.

This ability to adapt is what makes the V-Pots so powerful.

### Encoders Rather Than Pots

Despite the name V-Pot, these controls are not conventional potentiometers.

A traditional potentiometer has physical end stops. Turn it fully anticlockwise and eventually it can go no further. The physical position of the knob therefore represents the current value.

The X-Touch V-Pots work differently.

They are endless rotary encoders.

You can continue turning them in either direction indefinitely.

This matters because the same physical encoder can control many different parameters without its position becoming meaningless when you change from one parameter to another.

### Field Note

The word V-Pot is inherited from the Mackie Control terminology.

In practical terms, think of it as a pushable endless rotary encoder surrounded by an LED display.

## Why Endless Encoders Matter

Imagine that an ordinary knob is controlling pan and is currently turned fully to the left.

You then switch modes and assign that same knob to a plug-in parameter whose current value is 80%.

What should happen?

With a conventional potentiometer there is an immediate disagreement between the physical position of the knob and the value in the software.

An endless encoder has no such problem.

The X-Touch simply changes the LED ring to show the new value.

The encoder is immediately ready to adjust it.

This is another example of the controller adapting itself to the current context.

## The LED Ring

Each V-Pot is surrounded by a ring of LEDs.

As we saw in the previous chapter, this is an important part of the X-Touch's feedback system.

The LEDs provide a visual indication of the value currently assigned to the encoder.

For a pan control, for example, the display can indicate whether the signal is positioned:

- left
- centre
- right

For other parameters the LEDs may instead represent a value progressing from minimum to maximum.

The important point is that the LED ring represents the current software value, not a permanent physical position of the encoder.

### Reality Check

When you change modes and the LED rings suddenly change, nothing has gone wrong.

The encoders are simply displaying the values of the parameters they now control.

## Turning a V-Pot

Turning an encoder changes the currently assigned parameter.

Which parameter that happens to be depends upon context.

For example, the V-Pots may control:

- track pan
- send levels
- device parameters
- instrument parameters
- plug-in parameters
- browser navigation

You therefore cannot answer the question:

"What does this V-Pot control?"

without first answering another question:

"What mode am I in?"

This is exactly the mental model introduced earlier in the book.

The hardware remains the same.

Its meaning changes.

## Pressing a V-Pot

The V-Pots can also be pressed.

This is easy to overlook when first using the X-Touch because they look primarily like rotary controls.

The push action gives DrivenByMoss another command that can be associated with each encoder.

Its precise function depends upon the current context.

In one mode it may perform an action associated with the parameter being displayed; in another, it may be used for navigation or selection.

Do not assume that pressing an encoder always performs the same operation. Instead, look at the current mode and the information displayed on the scribble strips.

#### Field Note

The V-Pot's push action is one of the reasons it is useful to think of the encoder and the scribble strip beneath it as a pair.

The display tells you the context.

The V-Pot lets you interact with it.

## V-Pots and Scribble Strips

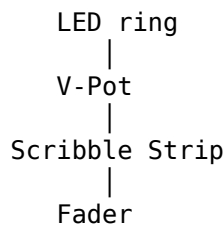
The relationship between the V-Pots and the scribble strips is fundamental.

Each encoder sits directly above a display.

That physical arrangement is deliberate.

When a mode assigns parameters to the encoders, the corresponding scribble strips can tell you what those parameters are.

Conceptually, each channel becomes:



The V-Pot provides the input.

The LED ring shows the value.

The scribble strip provides the context.

Together they form a small control surface within the larger controller.

## Pan Control

One of the simplest uses of the V-Pots is track panning.

When the controller is operating in the appropriate pan mode, each encoder controls the pan position of its corresponding channel.

Turn left and the signal moves towards the left.

Turn right and it moves towards the right.



The LED ring provides immediate visual feedback.

This is a good mode in which to become familiar with the physical feel of the encoders because the relationship between movement and result is easy to understand.

## Send Control

Switch to SEND mode and the same eight encoders take on a different role.

Instead of controlling pan, they can now adjust send levels.

The physical hardware has not changed.

Only the meaning assigned to it has changed.

This is precisely why endless encoders are so well suited to a controller such as the X-Touch.

The LED rings simply update to show the new values.

## Device and Plug-In Control

The real power of the V-Pots becomes apparent when controlling devices.

In Device or PLUG-IN mode, the encoders can be mapped to parameters belonging to the currently selected device.

Instead of:

Pan   Pan   Pan   Pan   Pan   Pan   Pan   Pan

you might see parameters representing:

Cutoff   Reso   Attack   Decay   Sustain   Release   Drive   Mix

The scribble strips identify the parameters.

The LED rings show their current values.

The V-Pots adjust them.

Suddenly the X-Touch becomes much more than a mixer.

It becomes a hands-on device editor.

## One Encoder, Many Jobs

This is worth emphasising.

The first V-Pot is not permanently:

Pan for Track 1

Nor is it permanently:

Parameter 1

It is simply the first encoder in the current context.

Change bank and it may control another track.

Change mode and it may control another type of parameter.

Select another device and it may acquire another purpose entirely.

Understanding this prevents one of the most common sources of confusion when learning a control surface.

## Turn, Look, Listen

When adjusting a V-Pot, you have several sources of information available at once.

You can:

- feel the encoder movement
- watch the LED ring
- read the scribble strip
- observe the corresponding change in Bitwig
- hear the result

That combination makes the V-Pots particularly effective for tasks such as sound design.

Instead of concentrating entirely on a graphical plug-in interface, you can begin to adjust parameters by ear.

### Field Note

Sometimes the best display in a music production system is no display at all.

Once you know which parameter a V-Pot controls, try listening to the result rather than watching the computer screen.

## V-Pots and Focus

The previous chapters introduced two important questions:

"What mode am I in?"

and:

"Which track is selected?"

V-Pots add a third:

"What parameter is this encoder controlling?"

Fortunately, you normally do not need to remember the answer.

The X-Touch tells you.

Read the scribble strip.

Look at the LED ring.

Then make the adjustment.

Once again:

Observe before you adjust.

## The Bigger Picture

The V-Pots demonstrate almost everything that makes the X-Touch different from a simple MIDI controller.

They are:

- context-sensitive
- bidirectional
- multifunctional
- visually informative
- integrated with the current selection and mode

A single row of eight encoders can therefore control a remarkable amount of Bitwig.

The trick is not to memorise every possible assignment.

The trick is to understand the context.

Once you do that, the V-Pots become some of the most intuitive controls on the entire surface.

## Exercise

Open a Bitwig project containing several tracks and devices.

Begin with the V-Pots controlling pan.

Turn several encoders and observe both the LED rings and Bitwig.

Next, switch to SEND mode.

Notice how the LED rings immediately change to represent the send values.

Finally, select a track containing a device and enter the appropriate device or plug-in mode.

Read the parameter names shown on the scribble strips and adjust several of them using the V-Pots.

Throughout the exercise, keep asking three questions:

1. Which track is selected?
2. Which mode am I in?
3. What parameter does this V-Pot currently represent?

Do not try to memorise the assignments.

Instead, practise reading the controller until the answer becomes obvious from the feedback it provides.

# Project XTC

## X-Touch Companion

The Unofficial Guide to the Behringer X-Touch, Bitwig Studio and DrivenByMoss

## Chapter 10 - Motor Faders

The nine motorised faders are probably the first thing most people notice about the X-Touch.

Eight belong to the channel strips, while the ninth sits separately as the master fader.

Move a fader and Bitwig responds.

Change a value in Bitwig and the fader moves by itself.

Switch banks and all eight channel faders can spring into new positions almost simultaneously.

It is impressive to watch, but the motors are not there simply for show.

They solve one of the fundamental problems of using physical controls with software.

### The Problem with Ordinary Faders

Imagine a simple MIDI controller with eight conventional faders.

The first fader is controlling Track 1, whose volume is currently fairly high.

Now switch to another bank.

That same physical fader might now control Track 9, whose volume is much lower.

The hardware fader is still sitting near the top of its travel, while the software value it represents is somewhere near the bottom.

The two disagree.

Different controllers solve this problem in different ways, often using techniques such as value pickup or soft takeover.

The X-Touch takes a much more direct approach.

It simply moves the fader.

### Field Note

Motorisation means that the physical position of a fader can always reflect the value currently held by Bitwig.

The hardware adapts to the software rather than asking you to compensate for the difference.

## Faders as Input and Output

It is tempting to think of a fader purely as an input device.

You move it.

Bitwig receives a new value.

But an X-Touch motor fader works in both directions.

You move the fader

↓  
Bitwig

Bitwig  
↓

The fader moves

This makes the fader both a control and a display.

Its position is information.

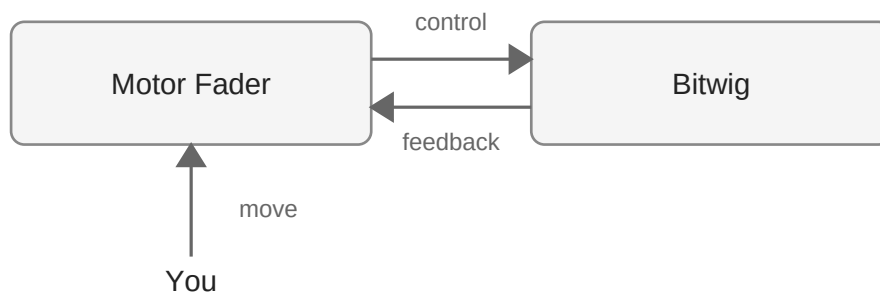


Figure 9.1 — The motor fader is both an input control and a physical display of Bitwig's current value.

## Eight Channel Faders

The eight main faders correspond to the eight channels in the current bank.

In a simple project this might mean:

|       |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|
| Fader | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| Track | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

Move to the next bank and those same physical faders might represent:

|       |   |    |    |    |    |    |    |    |
|-------|---|----|----|----|----|----|----|----|
| Fader | 1 | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| Track | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 |

As soon as the bank changes, the motors reposition the faders to show the values belonging to the new tracks.

This is the physical manifestation of the banking system introduced in Chapter 4.

## Banking Makes Sense of the Motors

Without motorisation, changing banks would create an immediate mismatch between the physical faders and the newly displayed tracks.

With motorised faders, the transition is obvious.

Press a bank control and three things happen together:

- the visible tracks change
- the scribble strips update
- the faders move to their new positions

The X-Touch has physically reconfigured itself for the new bank.

### Reality Check

If several faders suddenly move when you change banks, nothing unexpected has happened.

They are simply taking up the positions belonging to the newly displayed tracks.

## Touch-Sensitive Faders

The X-Touch faders are not merely motorised.

They are also touch-sensitive.

The controller can detect when your finger is resting on a fader.

This is particularly important when working with automation.

There is a significant difference between:

"The fader is at this position."

and:

"The user is currently touching the fader."

Knowing when you have physically taken control allows Bitwig and the controller to behave appropriately during automation operations.

We will encounter this distinction again when we look at automation in more detail.

## Do Not Fight the Motors

When a motor fader moves by itself, let it move.

Do not hold it in position or deliberately push against the motor.

The movement is feedback from Bitwig.

The fader is trying to show you the value that currently exists in the project.

If you want to change that value, wait for the fader to settle and then move it normally.

### Field Note

A moving fader is not the X-Touch doing something to your mix.

It is the X-Touch showing you something that has already happened in Bitwig.

That distinction is important.

## Automation Comes Alive

Motor faders become particularly valuable when automation is playing.

A track's volume may change continuously during a song.

On a conventional controller, the physical fader would remain stationary while the software value moved.

On the X-Touch, the motor fader can follow those changes.

You can therefore see the automation physically happening.

A vocal may rise slightly during a chorus.

A pad may fade away.

An effects return may increase during a transition.



The corresponding faders can move as those changes occur.

This is one of the moments when the distinction between software and hardware begins to disappear.

## Writing Automation

Touch sensitivity and motorisation become especially useful when writing automation.

You can grab a fader and perform a level change naturally with your hand rather than drawing an automation curve with a mouse.

Bitwig records the movement.

When the automation is played back, the fader can reproduce it.

The physical gesture has become part of the project.

For many users this feels much closer to working on a traditional mixing console.

## The Master Fader

To the right of the eight channel faders is a ninth motorised fader.

This is the master fader.

Unlike the channel faders, it is not part of the eight-channel banking system.

Its role is associated with the master output rather than whichever eight tracks happen to be visible.

This separation is useful because the master remains available while you move through the project's channel banks.

## Faders and SELECT

Remember the distinction introduced in Chapter 6.

A fader represents a track within the current bank.

The SELECT button identifies the track that currently has your attention.

Moving a fader does not mean that you have changed the controller's focus.

Likewise, selecting a track does not mean that its fader must move.

These are different concepts:

Fader position → Track level

SELECT → Current focus

Keeping those two ideas separate prevents a surprising amount of confusion.

## Faders as Feedback

Chapter 7 described the various ways in which the X-Touch communicates with you.

Motor faders are perhaps its most physical form of feedback.

You do not have to read a number to know roughly where a track's level is.

You can simply look at the fader.

In fact, you can often see the overall shape of a mix by looking across all eight faders at once.

That is something a row of numerical values on a computer screen does not communicate nearly as naturally.

## Mixing with Your Hands

This is where the X-Touch begins to change the experience of using a DAW.

With a mouse, adjusting several track levels usually means moving one virtual fader at a time.

With the X-Touch you have eight physical faders beneath your fingers.

You can bring one track down while raising another.

You can balance several channels together.

You can make small changes without repeatedly pointing at different objects on the screen.

Mixing becomes a physical activity again.

### Field Note

The advantage of physical faders is not simply that they replace virtual ones.

The real advantage is that you can control several things at once.

That is something a mouse is fundamentally bad at doing.

## Trust the Position

A useful habit is to trust what the faders are showing you.

When you change banks, allow them to settle.

When automation is playing, watch them move.

When you open an existing project, notice how the X-Touch reconstructs the mix physically in front of you.

The faders are not approximations of the software state.

They are part of the feedback system that keeps the controller and Bitwig synchronised.

## The Bigger Picture

The motor faders demonstrate the same principle we encountered with the V-Pots.

The X-Touch does not expect its physical controls to have one permanent meaning.

Instead, the hardware adapts itself to the current context.

The V-Pots achieve this by having no fixed physical position and using LED rings to display their values.

The faders achieve it differently.

They physically move.

Two different engineering solutions serve the same underlying idea:

The controller should reflect the current state of the software.

That principle lies at the heart of the X-Touch.

## Exercise

Open a Bitwig project containing more than eight tracks.

First, look at the positions of the eight channel faders.

Change to the next bank and watch what happens.

Notice how the scribble strips and faders update together.

Move several faders and confirm that the corresponding Bitwig levels follow them.

Now change those same levels using Bitwig's mixer and watch the X-Touch faders respond.

If the project contains volume automation, play a section containing that automation and observe the appropriate fader.

Finally, return to the first bank.

The faders should return to the positions belonging to those tracks.

The aim of this exercise is not simply to practise moving faders.

It is to experience the central idea of this chapter:

The motor faders are both controls and physical displays.

# Project XTC

## X-Touch Companion

The Unofficial Guide to the Behringer X-Touch, Bitwig Studio and DrivenByMoss

## Chapter 11 - Transport Controls

After learning about modes, selection, V-Pots and motor faders, the Transport section of the X-Touch may come as something of a relief.

Most of it does exactly what you expect.

The familiar REWIND, FAST FORWARD, STOP, PLAY and RECORD buttons provide immediate control over Bitwig's transport.

But the Transport section is more than a convenient replacement for clicking Play on the computer screen.

Combined with the jog wheel and navigation controls, it allows you to move around a project, locate a position and control playback without repeatedly reaching for the mouse.

### The Five Transport Buttons

At the bottom-right of the X-Touch are five large transport buttons:

- REWIND
- FAST FORWARD
- STOP
- PLAY
- RECORD

Their layout deliberately resembles the transport controls found on tape machines, hardware recorders and traditional mixing systems.

If you have used almost any audio equipment before, they should feel immediately familiar.

### PLAY

Press PLAY to start playback from the current position.

Pressing PLAY is one of the simplest examples of the two-way relationship between the X-Touch and Bitwig.

The controller sends the command.

Bitwig begins playback.

The X-Touch then reflects the resulting transport state through its illuminated controls and displays.

Even here, the controller is both issuing a command and providing feedback.

## STOP

Press STOP to stop playback.

Simple as this sounds, having a large physical STOP button becomes surprisingly valuable.

There is no need to locate a small transport control on the computer screen.

Your hand quickly learns where STOP is, and after a while you may find yourself using it without looking at the controller at all.

### Field Note

Transport controls are particularly good candidates for developing muscle memory.

Unlike context-sensitive controls, their basic purpose remains predictable.

Once your hand knows where PLAY and STOP are, controlling playback becomes almost unconscious.

## RECORD

The RECORD button controls Bitwig's recording transport.

Recording involves more than simply pressing one button — tracks must be armed correctly and Bitwig must be ready to record — but once those conditions are satisfied, RECORD gives you direct physical control over the process.

This becomes especially useful when recording instruments or automation.

Instead of preparing everything and then reaching for the mouse, you can remain focused on the performance.

### Reality Check

Pressing RECORD does not automatically make every track record.

Track arming and the transport's recording state are separate concepts.

Always check that the intended track is correctly armed before beginning a take.

## REWIND and FAST FORWARD

The REWIND and FAST FORWARD buttons allow you to move backwards and forwards through the project.

Their exact behaviour is governed by the integration between the X-Touch, DrivenByMoss and Bitwig, but their purpose is straightforward:

Move the current playback position.

For coarse navigation they provide a quick way of moving through the project without touching the mouse.

For more deliberate positioning, however, another control becomes particularly useful.

## The Jog Wheel

The large wheel above the transport buttons is the jog wheel.

It provides a physical way of moving through the project timeline.

Turn it one way to move backwards.

Turn it the other way to move forwards.

Unlike clicking somewhere on a graphical timeline, the jog wheel encourages you to think in terms of movement from the current position.

This can feel much more natural when searching for the beginning of a phrase, a transition or a particular point in a recording.

## Coarse and Fine Navigation

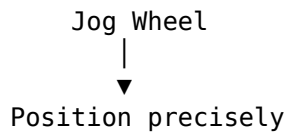
The transport buttons and jog wheel complement each other.

Think of them broadly as two approaches to navigation:

REWIND / FAST FORWARD



Move quickly



You do not need to choose one method exclusively.

A typical workflow might involve moving rapidly towards the required part of the project and then using the jog wheel to refine the position.

#### Field Note

The most useful navigation method is usually the one that requires the least thought.

Use the transport buttons when you want to move.

Use the jog wheel when you want to arrive somewhere specific.

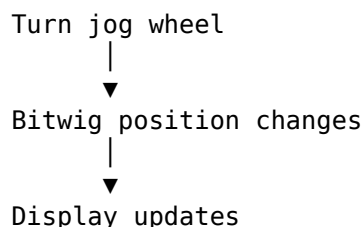
## The Time Display

Navigation becomes much more useful when you can see where you are.

The X-Touch's main display provides positional information that helps you keep track of the current location within the project.

As you move through the timeline, the display updates accordingly.

Once again, input and feedback work together:



You move.

The controller tells you where you have arrived.

## Transport and the Computer Screen

It is worth noticing how many routine mouse movements disappear once the transport controls become familiar.

Without a control surface, a typical editing or recording session may involve repeatedly moving the pointer between:

- the arrangement
- the mixer



- device controls
- the transport
- the timeline

With the X-Touch, many transport operations remain permanently beneath your hand.

You can keep the computer screen focused on the information that actually needs a graphical display.

## Transport While Mixing

Physical transport controls are especially useful while mixing.

Imagine adjusting several faders during playback.

You hear something you want to investigate.

Instead of moving your hand away from the controller and locating Bitwig's transport:

1. Press STOP.
2. Navigate backwards.
3. Press PLAY.
4. Listen again.
5. Continue adjusting the mix.

Your attention remains on the sound and the controller rather than shifting repeatedly between hardware and screen.

This may seem like a small improvement.

Repeated hundreds of times during a session, it becomes a substantial one.

## Transport While Recording

The same principle applies when recording.

A typical sequence might be:

1. Select the required track.
2. Arm it for recording.
3. Navigate to the starting position.
4. Press RECORD.
5. Perform the part.
6. Press STOP.
7. Return to the beginning of the take.
8. Press PLAY to review it.

The computer is still doing all the recording.

The X-Touch simply allows the routine operations surrounding that recording to happen physically.

That keeps your attention where it belongs: on the performance.

## Visual Feedback

The transport section also reinforces the principle introduced in Chapter 7:

Observe before you adjust.

Illuminated transport buttons tell you about the current state.

The position display tells you where you are.

Bitwig's own interface provides further confirmation.

Do not treat the physical controls and the software interface as two independent systems.

They are two views of the same transport state.

## Developing Muscle Memory

Transport is one of the best places to begin operating the X-Touch without constantly looking at it.

The five transport buttons are large, consistently positioned and easy to distinguish by location.

With practice, your hand begins to find them automatically.

The jog wheel is equally unmistakable.

This is an important step towards a more tactile workflow.

You stop thinking:

"Where is the Play button?"

and simply press it.

That small change is one of the reasons a control surface can make a DAW feel more like an instrument and less like a computer application.

## A Step Towards Mouse-Lite

The Transport section gives us an early glimpse of the workflow we shall explore more fully in Chapter 13.

The aim is not necessarily to eliminate the mouse.

There are many tasks for which a mouse remains an excellent tool.

The aim is to stop reaching for it when a physical control is faster, clearer or more natural.

Starting playback is a perfect example.

You could move the pointer to Bitwig's PLAY button and click it.

But when a large physical PLAY button is already beneath your hand, why would you?

#### Field Note

A Mouse-Lite workflow is not about refusing to use the mouse.

It is about using the most appropriate tool for each job.

For transport operations, that tool is very often the X-Touch.

## The Bigger Picture

The Transport section may be one of the least mysterious parts of the X-Touch, but it plays an important role in the overall workflow.

The faders allow you to mix.

The V-Pots allow you to adjust parameters.

SELECT establishes focus.

The displays tell you what is happening.

And the Transport section lets you control when it happens.

Together, these controls begin to form a complete working environment.

You are no longer simply controlling individual Bitwig parameters.

You are operating the session.

## Exercise

Open an existing Bitwig project.

For the duration of this exercise, avoid using the mouse for transport operations.

Using only the X-Touch:

1. Start playback.
2. Stop playback.
3. Move backwards through the project.
4. Move forwards through the project.
5. Use the jog wheel to locate a particular section.
6. Start playback from that position.
7. Stop again.

Now choose a short section of the project and practise repeatedly navigating to it and playing it.

Do not concentrate on speed.

Instead, concentrate on allowing your hand to learn where the transport controls are.

The goal is to reach the point where PLAY, STOP and basic navigation no longer require conscious thought.

That is one more step towards a Mouse-Free — or Mouse-Lite — workflow.

# Project XTC

## X-Touch Companion

The Unofficial Guide to the Behringer X-Touch, Bitwig Studio and DrivenByMoss

## Chapter 12 - Device Mode

So far, much of our attention has been focused on the mixer.

We have selected tracks, changed banks, adjusted levels, controlled pan and navigated through the project.

But a Bitwig track is much more than a channel in a mixer.

It may contain instruments, audio effects and other devices, each with parameters of its own.

Device Mode is where the X-Touch begins to reach inside the selected track and give you physical control over those devices.

### From Mixer to Device

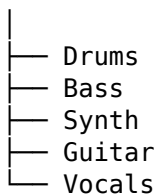
In normal mixer operation, the eight channel strips represent tracks.

Conceptually, the controller looks something like this:

X-Touch



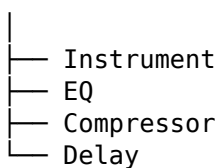
Tracks



Device Mode takes us one level deeper.

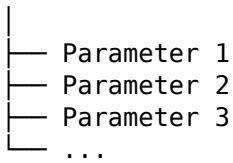
A selected track might contain:

Synth Track



Once a device is selected, we can go deeper again:

Selected Device



This hierarchy is one of the most useful ways to understand Device Mode.

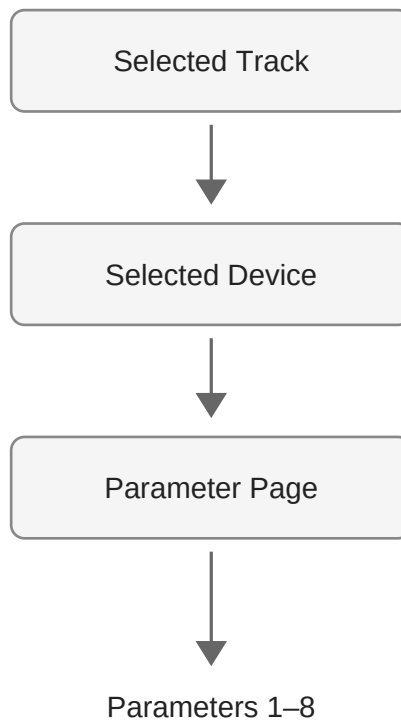


Figure 11.1 — Device Mode moves the controller's focus from the selected track, through its devices, to the parameters of the selected device.

## SELECT Comes First

Chapter 6 introduced the idea of focus.

Device Mode is where that idea becomes particularly important.

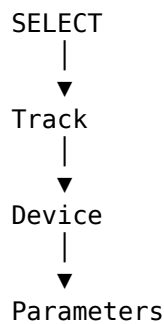
Before controlling a device, the X-Touch needs to know which track contains it.

The first step is therefore familiar:

Select the track.

Once the track is selected, Device Mode can operate on the devices belonging to that track.

This gives us a simple hierarchy:



If Device Mode appears to be showing the wrong instrument or effect, work backwards through that hierarchy.

Start by checking the selected track.

#### Reality Check

When the X-Touch appears to be controlling the "wrong" device, do not immediately start pressing mode buttons.

First check which track is selected.

The controller may be doing exactly what it should — just on a different track.

## A Track Can Contain Many Devices

A Bitwig track can contain an entire chain of devices.

For example:

Polysynth → EQ+ → Compressor → Delay+

Each device has a different purpose and therefore exposes different parameters.

Selecting the Polysynth might give you controls relating to synthesis.

Selecting the EQ might give you frequency and gain controls.

Selecting the delay might expose delay time, feedback and mix.

The same eight V-Pots can control all of them.

The hardware remains unchanged.

The context changes.

## The Scribble Strips Become Essential

This is where the scribble strips really earn their keep.

In mixer operation, an encoder's purpose may be fairly predictable.

In Device Mode, it could represent almost anything.

One encoder might control:

- filter cutoff
- resonance
- attack
- release
- frequency
- gain
- feedback
- wet/dry mix

Trying to memorise those assignments would be hopeless.

Fortunately, you do not need to.

The scribble strips tell you what the controls currently represent.

This is precisely the behaviour we explored in Chapter 7.

Observe before you adjust.

In Device Mode, that principle becomes essential.

## The V-Pots Become Parameter Controls

Chapter 8 introduced the V-Pots as context-sensitive rotary encoders.

Device Mode demonstrates why that design is so useful.

Suppose the currently selected device exposes eight parameters:

Cutoff   Reso   Attack   Decay   Sustain   Release   Drive   Mix

The eight V-Pots can immediately become physical controls for those parameters.

Turn an encoder and the corresponding value changes in Bitwig.

The LED ring provides visual feedback.

The scribble strip tells you what the parameter is.

Suddenly, a software device begins to feel rather more like a piece of hardware.

## More Than Eight Parameters

Of course, most devices have considerably more than eight parameters.

The X-Touch only has eight V-Pots.



This creates a familiar problem.

Fortunately, we have already encountered the solution.

Banking.

Just as eight channel strips can provide access to many tracks, eight encoders can provide access to many parameters.

Think of the device parameters as pages:

Page 1

P1   P2   P3   P4   P5   P6   P7   P8

Page 2

P9   P10   P11   P12   P13   P14   P15   P16

Page 3

P17   P18   P19   P20   P21   P22   P23   P24

Move to another parameter page and the V-Pots acquire new assignments.

The scribble strips update.

The LED rings update.

Once again, the controller reconfigures itself around the current context.

#### Field Note

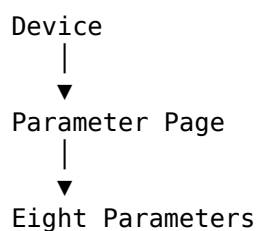
Parameter pages are really just another form of banking.

You already understand the underlying idea from Chapter 4.

Eight physical controls provide a window onto a much larger collection of software controls.

## Devices, Parameters and Pages

It is useful to keep three separate concepts in mind:



First you choose what device you want to control.

Then you choose which page of its parameters you want to see.

The eight V-Pots then control the parameters on that page.

This structure may sound complicated when described in words.

In use, it quickly becomes natural because the X-Touch continually updates its displays as you navigate.

## Follow the Feedback

Imagine moving from a synthesizer to a delay.

The controller may immediately change:

- parameter names on the scribble strips
- LED ring positions
- encoder functions

Nothing has physically moved except perhaps your hand.

Yet the entire row of V-Pots now represents a different piece of software.

This is exactly why Chapter 7 placed so much emphasis on feedback.

In Device Mode, the displays are not merely helpful.

They are part of the interface.

## Editing by Ear

One of the most enjoyable aspects of Device Mode is that it can change how you interact with plug-ins and instruments.

Consider adjusting a filter cutoff with a mouse.

Your attention naturally goes to the computer screen.

You locate the graphical control.

You watch the pointer.

You watch the value change.

With the X-Touch, you can instead:

1. Read the scribble strip.
2. Place your hand on the appropriate V-Pot.
3. Turn it.
4. Listen.

Once you know which control you are adjusting, there may be very little reason to keep watching the screen.

### Field Note

Device Mode is not necessarily about reproducing every control from a plug-in interface on the X-Touch.

Its real value is providing immediate physical access to the parameters you actually want to adjust.

## Hardware Feel from Software Instruments

This becomes particularly interesting with software instruments.

A synthesizer may exist entirely inside the computer, but Device Mode gives some of its controls a physical presence.

Instead of dragging a virtual filter knob, you turn an encoder.

Instead of watching an envelope value, you adjust it while listening.

The software has not changed.

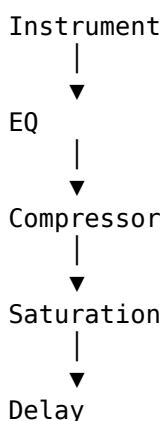
What has changed is your relationship with it.

This is one of the ways in which a control surface can make working with a DAW feel less like operating software and more like playing an instrument.

## Device Chains

As projects become more complex, tracks often contain several devices.

For example:



Device Mode allows you to move through that chain and concentrate on one device at a time.

The important thing is not to memorise where every device lives.

Instead, maintain the mental hierarchy:

Track → Device → Parameter Page → Parameter

Whenever you become lost, move back up that hierarchy.

Which track?

Which device?

Which page?

Which parameter?

The displays provide the answers.

## Device Mode and Focus

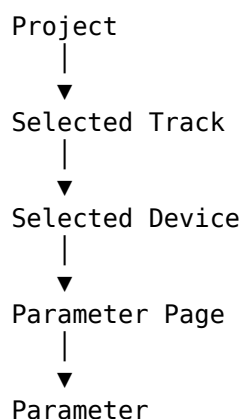
We can now expand the idea introduced in Chapter 6.

SELECT establishes the track focus.

Device Mode establishes the device focus.

Parameter navigation establishes the parameter focus.

Conceptually:



This is not just a description of Device Mode.

It is a useful mental model for navigating Bitwig itself.

## Returning to the Mixer

Device Mode is temporary.

Entering it does not alter the structure of your project or permanently reassign the X-Touch.

When you have finished editing the device, you can return to normal mixer operation.

The V-Pots resume their mixer-related roles.

The scribble strips return to displaying the appropriate mixer information.

The controller has not changed identity.

It has simply changed context again.

## Device Mode and the Mouse

Device Mode also represents another significant step towards our Mouse-Free — or Mouse-Lite — workflow.

Without a control surface, editing a device often involves:

- locating the device on screen
- locating the required parameter
- moving the pointer to it
- dragging the control
- moving to the next parameter
- repeating the process

With the X-Touch, once the correct device and parameter page are selected, eight physical controls are immediately available.

That does not mean the graphical interface becomes useless.

Some devices contain complex visual information that is much easier to understand on screen.

The point is not to eliminate the screen or the mouse.

The point is to stop depending upon them when a physical control provides a better way to work.

### Field Note

Mouse-Lite does not mean "never touch the mouse".

It means:

Use the mouse when it is the best tool — not simply because it is the only tool you know.

## The Bigger Picture

Device Mode brings together almost every concept we have encountered so far.

Banks and Channels taught us how a limited number of physical controls can represent a larger system.

Modes taught us that controls change meaning according to context.

SELECT taught us about focus.

Displays and Feedback taught us to read the controller.

V-Pots gave us context-sensitive parameter controls.

Device Mode combines all of those ideas.

That is why it can initially appear complicated.

It is also why, once the earlier concepts are understood, it becomes surprisingly logical.

You do not need to memorise the entire X-Touch.

You simply need to know where you are.

## Exercise

Open a Bitwig project containing a track with several devices.

Select that track using the X-Touch.

Enter Device Mode and observe the scribble strips before touching any controls.

Identify the currently selected device.

Now:

1. Locate a parameter shown on one of the scribble strips.
2. Adjust it with the corresponding V-Pot.
3. Watch the LED ring respond.
4. Listen to the result.
5. Move to another parameter page if one is available.
6. Observe how all eight encoder assignments change.
7. Select another device in the chain.
8. Observe how the scribble strips and LED rings change again.

Finally, select a completely different track.

Notice how the available devices change with it.

Throughout the exercise, keep the hierarchy in mind:

Track → Device → Parameter Page → Parameter

If you become lost, do not start pressing buttons at random.

Work backwards through the hierarchy and read the feedback the X-Touch is already providing.

# Project XTC

## X-Touch Companion

The Unofficial Guide to the Behringer X-Touch, Bitwig Studio and DrivenByMoss

## Chapter 13 - Browser Mode

Up to this point, most of our work with the X-Touch has involved controlling things that already exist in the project.

We have selected tracks.

We have adjusted levels.

We have navigated devices and changed their parameters.

But sooner or later we need something new.

Perhaps we want to add an instrument.

Perhaps we need an audio effect.

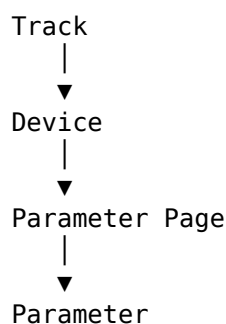
Perhaps we want to replace a preset or search for a particular sound.

This is where Browser Mode becomes particularly interesting.

Instead of reaching immediately for the mouse, we can begin exploring Bitwig's Browser directly from the X-Touch.

### From Control to Creation

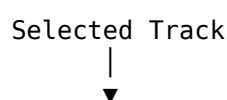
Device Mode allows us to move down through a hierarchy:



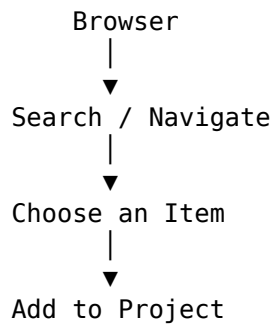
Browser Mode performs a different job.

It allows us to choose something that is not yet part of that hierarchy.

Conceptually:







This is an important change.

The X-Touch is no longer merely editing the project.

It is helping us build it.

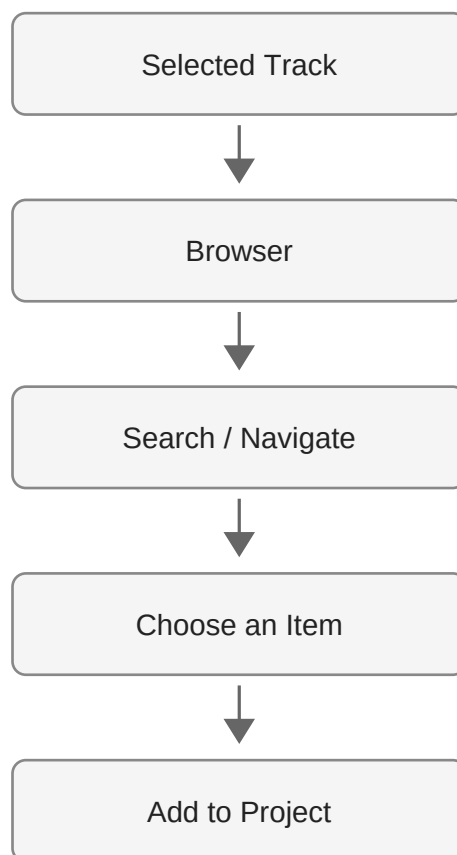


Figure 12.1 — Browser Mode extends the workflow from selecting a destination to finding and adding new material.

## SELECT Still Comes First

Once again, the SELECT button plays a central role.

Before opening the Browser, ask:

"Where do I want the new item to go?"

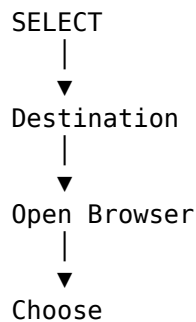
The selected track establishes the context in which browsing takes place.

This is the same idea introduced in Chapter 6.

Selection is not simply about highlighting something.

It establishes focus.

A useful working sequence is therefore:



That first step prevents a surprising amount of confusion.

#### Reality Check

Before browsing for a new device or sound, check the selected track.

Finding exactly what you wanted and then discovering that you were working in the wrong place is considerably less entertaining than it sounds.

## Opening the Browser

With DrivenByMoss, the X-Touch provides access to Bitwig's Browser without requiring you to begin the operation with the mouse.

When Browser Mode is entered, the role of the controller changes.

Controls that previously represented mixer or device parameters can now become navigation and selection controls.

The scribble strips change accordingly.

This should already feel familiar.

The hardware has not changed.

The context has.

## Read the Scribble Strips

Browser Mode is another excellent demonstration of why the scribble strips matter.

When browsing, they may present information relating to the available choices rather than track or device parameters.

The labels on the controller therefore become your guide.

Do not assume that an encoder still performs the function it had a few seconds earlier.

Read first.

Then turn or press.

The principle remains:

Observe before you adjust.

Although in Browser Mode we might reasonably amend that to:

Observe before you choose.

## Navigation Rather Than Parameter Control

In Device Mode, turning a V-Pot normally changes a value.

Browser Mode can give those same controls a different purpose.

Now an encoder may be involved in navigating choices rather than adjusting a continuous parameter.

This distinction is important.

In one context:

Turn V-Pot  
|  
▼  
Change Parameter

In another:

Turn V-Pot  
|  
▼  
Navigate Choices

The physical action is identical.

Its meaning is determined by the current mode.

This is precisely the mental model we established much earlier in the book.

## Narrowing the Search

Bitwig's Browser can contain a huge amount of material.

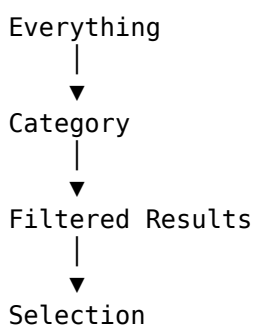
Depending upon your installation, that may include:

- instruments
- audio effects
- note effects
- presets
- samples
- plug-ins
- other device content

Browsing everything at once would quickly become unwieldy.

The useful approach is to narrow the available choices until the required item becomes easy to locate.

Think of browsing as a funnel:



Each decision reduces the number of possibilities.

This is much more effective than treating the Browser as one enormous list.

## Browsing Is Contextual

The contents of the Browser depend upon what you are doing.

If you are adding an instrument, the useful choices are different from those involved in adding an audio effect.

If you are choosing a preset, the available material depends upon the device.

If you are looking for a sample, you are dealing with another type of content again.

This means Browser Mode should not be thought of as:

"A list of everything Bitwig contains."

It is better understood as:

"A way of choosing something appropriate for the current context."

That makes the Browser much less intimidating.

## Choosing an Item

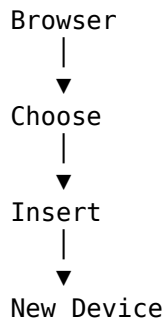
Eventually, browsing leads to a choice.

Once the required item has been located, it can be selected and inserted or loaded in the appropriate context.

At this point the Browser has done its job.

The new item becomes part of the project.

Conceptually:



And something rather satisfying can happen next.

The workflow can move naturally back into Device Mode.

## Browser Mode Meets Device Mode

Suppose you want to add a delay to a track.

The workflow might be:

1. SELECT the destination track.
2. Open the Browser.
3. Navigate to an appropriate delay.
4. Choose it.
5. Return to Device Mode.
6. Adjust its parameters with the V-Pots.

Notice what has happened.

You have gone from:

"I want a delay."

to:

"I am adjusting the delay."

without the mouse necessarily being the centre of the operation.

This is where the individual chapters of the book begin joining together into complete workflows.

### Field Note

Browser Mode becomes much more powerful when you stop thinking of it as an isolated feature.

It is the bridge between deciding that you need something and then controlling what you have added.

## Browsing for Instruments

The same principle applies when building an instrument track.

Imagine starting with an appropriate empty track.

You can:

1. Select the track.
2. Enter the Browser.
3. Locate an instrument.
4. Choose it.
5. Enter Device Mode.
6. Begin adjusting its parameters.

The Browser finds the instrument.

Device Mode controls it.

SELECT establishes where the whole operation takes place.

Three apparently separate features become a single workflow.

## Browsing for Presets

Browser Mode is equally useful when searching for sounds rather than devices.

Suppose you already have a synthesizer loaded but want another preset.

Instead of thinking:

"I need to operate the Browser."

think:

"I want another sound for this device."

The Browser is simply the mechanism that helps you find it.

This distinction matters because it keeps your attention on the musical task rather than the software operation.

## Browsing with Your Ears

Browsing sounds presents an interesting opportunity.

Computer-based browsing naturally encourages us to look.

We read names.

We study categories.

We watch lists move up and down the screen.

But ultimately, when choosing a sound, the important question is usually:

"Does this sound right?"

The more navigation you can perform from the controller, the easier it becomes to keep your attention on listening rather than pointing and clicking.

### Field Note

Names and categories help you find sounds.

Your ears decide whether they belong in the music.

## The Screen Still Matters

A Mouse-Lite workflow does not mean pretending the computer display has suddenly become useless.

Bitwig's Browser can present detailed information that simply cannot fit onto the X-Touch's scribble strips.

There will be occasions when looking at the screen is clearly the most efficient way to understand what is available.

That is perfectly fine.

The aim is not:

"Never look at Bitwig."

The aim is:

"Do not reach for the mouse automatically."

If the X-Touch can perform the operation comfortably, use it.

If the graphical Browser is better suited to the task, use that.

The two approaches complement each other.

## Do Not Memorise the Browser

As with Device Mode, there is little value in trying to memorise every possible Browser state.

The available choices will change.

Your installed devices may change.

Your plug-ins may change.

Your presets and samples may change.

The useful skill is not remembering where everything lives.

It is understanding how to navigate what is currently in front of you.

Read the feedback.

Make a choice.

Observe the result.

Continue.

## When You Become Lost

Browser Mode can initially feel more complicated than the mixer because the controls are being used in a less familiar way.

If you become unsure what is happening, return to the questions we have used throughout the book:

- Which track is selected?
- Which mode am I in?
- What are the scribble strips showing?
- What does the current control represent?

Do not press controls randomly in an attempt to escape.

Read the controller first.

The information you need is often already there.

### Reality Check

When a control appears to have stopped doing its usual job, check whether you are still in Browser Mode.

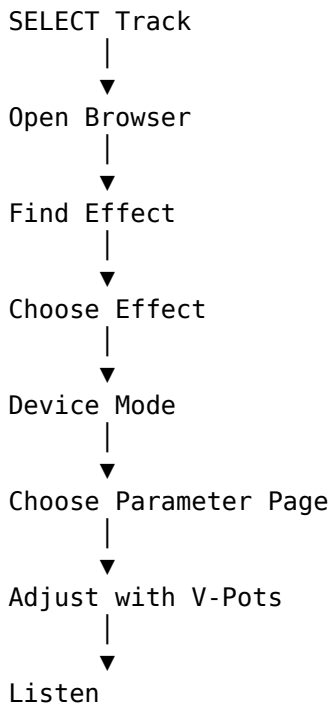
A V-Pot that normally adjusts pan cannot adjust pan while it is busy helping you navigate the Browser.



## Building a Workflow

We can now combine much of what we have learned into a single sequence.

Imagine adding and adjusting a new effect:



This is more than a collection of X-Touch features.

It is a workflow.

That distinction is important.

The purpose of learning the controller is not to become good at pressing its buttons.

The purpose is to make the controller disappear into the process of making music.

## Another Step Towards Mouse-Lite

Browser Mode brings us very close to the subject of the final chapter.

Earlier we used the X-Touch to manipulate a project.

Now we can also begin adding things to it.

That removes another common reason for immediately reaching for the mouse.

Again, the goal is not mouse avoidance for its own sake.

A mouse remains extremely useful for detailed editing, graphical operations and many other tasks.

But opening a Browser, finding something, selecting it and then adjusting it are operations that can increasingly take place from the control surface.

The mouse becomes an option rather than a reflex.

## The Bigger Picture

Browser Mode completes an important chain of ideas.

SELECT establishes where we are working.

Browser Mode helps us find something to add.

Device Mode lets us control what we have added.

V-Pots provide the physical controls.

Displays and Feedback tell us what those controls currently mean.

Motor Faders and Transport allow the rest of the session to continue around us.

The individual features of the X-Touch are beginning to disappear.

In their place we are developing workflows.

And that brings us naturally to the final chapter.

## Exercise

Open a Bitwig project containing a suitable track.

For this exercise, resist the temptation to begin with the mouse.

First:

1. Select the destination track from the X-Touch.
2. Enter Browser Mode.
3. Observe the scribble strips before touching anything else.
4. Navigate through the available choices.
5. Choose an instrument, effect or other suitable item.
6. Confirm that it has been added in the intended location.

Now move directly into Device Mode.

7. Identify the newly added device.
8. Locate a useful parameter.
9. Adjust it using a V-Pot.
10. Listen to the result.

Finally, return to normal mixer operation.

The purpose of this exercise is not to prove that every operation can be performed without a mouse.

It is to experience the complete sequence:

Select → Browse → Choose → Control → Listen

When that sequence begins to feel like a single operation rather than five separate features, the X-Touch is becoming part of your workflow.

# Project XTC

## X-Touch Companion

The Unofficial Guide to the Behringer X-Touch, Bitwig Studio and DrivenByMoss

## Chapter 14 - Mixer Edit Modes

### Mixer Edit Modes

By now, we have become used to an important characteristic of the X-Touch:

The physical controls stay in the same place, but what they control can change.

The eight channel strips normally give us a familiar view of the mixer.

We have faders for track levels, V-Pots for parameters, and buttons for operations such as SELECT, MUTE, SOLO and ARM.

DrivenByMoss can take this idea further.

Instead of thinking only in terms of eight complete channel strips, we can temporarily ask the X-Touch to concentrate on one particular aspect of the mixer.

That is the idea behind the mixer edit modes.

### Looking Across the Mixer

Imagine eight tracks:

Track 1   Track 2   Track 3   Track 4   Track 5   Track 6   Track 7   Track 8

There are several different ways we might want to look across them.

We could ask:

What are their volumes?

or:

What are their panorama positions?

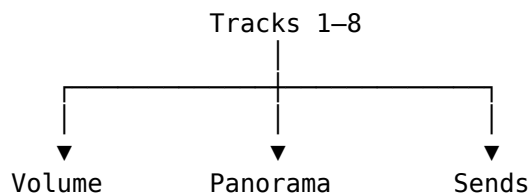
or:

How much of each track is being sent to Send 1?

The tracks have not changed.

What changes is the dimension of the mixer that we are looking at.

Conceptually:



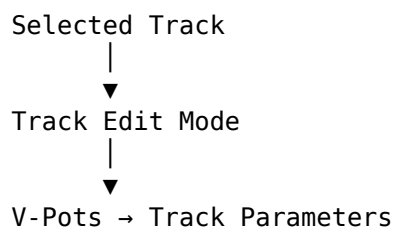
This is a useful way to understand the mixer edit modes.

## Track Edit Mode

Press TRACK to enter Track Edit Mode.

Rather than treating each V-Pot simply as the normal control for its channel strip, Track Edit Mode exposes parameters associated with the selected track.

This gives the V-Pots another role:



The precise parameters available can depend on the current DrivenByMoss configuration.

The important point is the change in perspective.

We are no longer primarily looking across eight tracks.

We are looking more deeply at one selected track.

## Volume Edit Mode

Press TRACK again and DrivenByMoss switches to Volume Edit Mode.

Now the V-Pots provide another way of controlling the volumes of the eight tracks in the current bank.

Conceptually:

V-Pot 1 → Track 1 Volume  
 V-Pot 2 → Track 2 Volume  
 V-Pot 3 → Track 3 Volume  
 ...  
 V-Pot 8 → Track 8 Volume

Of course, the X-Touch already has eight motor faders.

So why would we want volume on the V-Pots as well?

One answer is FLIP, which we met earlier.

The ability to exchange the roles of faders and rotary controls becomes much more useful when the controller can present parameters in different ways.

The important lesson is not that one control is the "correct" way to adjust volume.

It is that DrivenByMoss allows the surface to be reorganised around the task.

## Panorama Edit Mode

Press PAN to enter Panorama Edit Mode.

The eight V-Pots now control the panorama positions of the eight tracks in the current bank:

V-Pot 1 → Track 1 Pan  
V-Pot 2 → Track 2 Pan  
V-Pot 3 → Track 3 Pan  
...  
V-Pot 8 → Track 8 Pan

This is perhaps the clearest example of an edit mode.

Instead of thinking:

Channel 1  
├── Volume  
├── Pan  
├── Mute  
└── Solo

we temporarily rotate our view through ninety degrees and think:

Pan  
├── Track 1  
├── Track 2  
├── Track 3  
├── ...  
└── Track 8

The information is the same.

The view onto it has changed.

## Fine Adjustment with SHIFT

As we saw in Chapter 8, SHIFT frequently provides finer adjustment of continuous parameters.

That pattern applies here too.

Hold SHIFT while turning a V-Pot when you need more precise control.

Conceptually:

Turn V-Pot

↓  
Normal adjustment

SHIFT + Turn V-Pot

↓  
Finer adjustment

This is a good example of why we introduced modifiers before exploring the advanced modes.

The modifier itself is no longer something new to learn.

We simply apply an existing idea to a new task.

## Send Edit Modes

The SEND control opens another particularly useful family of mixer views.

Press SEND and the V-Pots control a Send across the eight tracks.

For example, in the first Send mode:

SEND 1

|         |   |        |
|---------|---|--------|
| Track 1 | → | Send 1 |
| Track 2 | → | Send 1 |
| Track 3 | → | Send 1 |
| Track 4 | → | Send 1 |
| Track 5 | → | Send 1 |
| Track 6 | → | Send 1 |
| Track 7 | → | Send 1 |
| Track 8 | → | Send 1 |

The eight V-Pots now answer one question:

How much of each track is being sent to this destination?

This is an especially natural way to work with effects sends.

If Send 1 feeds a reverb, for example, you can adjust the amount of reverb for eight tracks without changing the basic mixer bank.

## Moving Through the Sends

A project may contain more than one Send.

Repeated presses of SEND move through the available Send modes.

Conceptually:

SEND

↓

Send 1

↓

SEND

↓

Send 2  
↓  
SEND  
↓  
Send 3  
↓  
...

DrivenByMoss supports Send modes 1-8.

Hold SHIFT while pressing SEND to move through the Sends in the opposite direction.

So:

SEND  
→ next Send

SHIFT + SEND  
→ previous Send

This is another modifier pattern we have already encountered:

SHIFT can provide the related or reverse operation.

## Jumping Directly to a Send

Cycling is useful when moving between neighbouring Sends.

But if you already know which Send you want, DrivenByMoss provides a quicker route.

Hold SEND and use the channel SELECT buttons to choose the Send directly.

Conceptually:

SEND + SELECT 1 → Send 1  
SEND + SELECT 2 → Send 2  
SEND + SELECT 3 → Send 3  
...  
SEND + SELECT 8 → Send 8

This is an important example of a control acquiring a new meaning from context.

Normally:

SELECT 3

means:

Select Track 3.

But while SEND is being used as the context:

SEND + SELECT 3

means:



Select Send 3.

The physical SELECT button has not changed.

The context has.

## Switching a Send On or Off

Send level and Send state are two different things.

Turning a V-Pot adjusts the amount being sent.

DrivenByMoss also allows the Send itself to be switched on or off.

In Send Mode:

OPTION + V-Pot press

toggles the corresponding Send.

So one rotary control can provide both:

Turn

↓

Send level

OPTION + Press

↓

Send on/off

This is exactly the kind of compact interaction for which modifiers are useful.

The common operation remains immediately available.

The related secondary operation is one modifier away.

## Send Mode as a Mixing Tool

Send Mode is worth thinking about as more than a shortcut.

Suppose Send 1 contains a reverb and Send 2 contains a delay.

You can move between two complete views of your mix:

Send 1 – Reverb

---

Track 1 amount

Track 2 amount

Track 3 amount

...

Track 8 amount

then:

Send 2 – Delay

---

Track 1 amount

Track 2 amount  
Track 3 amount  
...  
Track 8 amount

Rather than visiting each track and adjusting one Send at a time, you can work across the mix according to the effect you are shaping.

For dub-style mixing in particular, this way of thinking can be extremely useful.

## ARM, MUTE and SOLO

The channel-strip buttons continue to provide immediate control over the tracks.

### ARM

Press ARM on a channel strip to arm that track for recording.

The bank of ARM buttons allows record state to be managed directly from the surface.

### MUTE

Press MUTE to mute the corresponding track.

### SOLO

Press SOLO to solo the corresponding track.

These are familiar mixer operations, but DrivenByMoss also provides modified versions for useful bank-wide or related operations.

## Clearing Mutes

When several tracks have been muted, clearing them individually can be tedious.

DrivenByMoss provides a global operation for clearing active mutes.

This is particularly useful after using mute creatively during playback.

Rather than searching the surface for every illuminated MUTE button, the controller can return the mixer to an unmuted state in one operation.

The exact global-control combination will be included in the Quick Reference once the final command set has been verified against the current DrivenByMoss version.

## Clearing Solos

The same problem occurs with Solo.

During mixing it is easy to accumulate a state in which one or more tracks are soloed.

DrivenByMoss provides a global operation for clearing active solos.

Again, the important workflow idea is:

Experiment

↓

Mute / Solo tracks

↓

Return the mixer to a known state

The controller is not merely a way to activate states.

It also provides efficient ways to recover from them.

## Monitoring

SHIFT-modified channel buttons provide access to monitoring-related functions.

These are useful when recording and will become more meaningful when we look at advanced recording workflows later in the guide.

For now, the important distinction is that ARM, MUTE and SOLO buttons are not necessarily limited to the words printed on their caps.

Like the rest of the X-Touch, they participate in the modifier system.

## FLIP

FLIP changes the relationship between rotary and fader control.

This becomes especially useful in the mixer edit modes.

A parameter that would normally be controlled from a V-Pot can be placed onto the motor faders.

Why might we want that?

Because a fader gives us:

- a long physical travel;
- touch sensitivity;
- motorised feedback;
- a very different physical feel from a rotary encoder.

For some adjustments, the V-Pot is ideal.

For others, the fader is more expressive.

FLIP lets the controller adapt.

Conceptually:

Before FLIP

V-Pots → parameter  
Faders → volume

After FLIP

V-Pots → volume  
Faders → parameter

The exact result depends on the current mode, but the principle remains the same:

FLIP exchanges control roles.

## Mixing by Dimension

The most useful way to think about this chapter is not as a collection of modes.

Think of it as several different ways to slice through the same mixer.

### By channel

Track 1  
├─ Volume  
├─ Pan  
├─ Sends  
├─ Mute  
└─ Solo

### By parameter

Pan  
├─ Track 1  
├─ Track 2  
├─ Track 3  
└─ ...

### By Send

Reverb Send  
├─ Track 1  
├─ Track 2  
├─ Track 3  
└─ ...

None of these is more correct than another.

They are different views of the same project.

## Choosing the Useful View

A good control surface should reduce the distance between an intention and an action.

If your intention is:

Turn Track 4 down.

the motor fader is probably already exactly what you want.

If your intention is:

Spread these eight tracks across the stereo field.

Panorama Mode gives you eight related controls together.

If your intention is:

Decide which tracks should feed the delay.

Send Mode gives you a complete view of that Send across the bank.

The important skill is therefore not memorising modes.

It is recognising which view best matches the job you are doing.

## The Important Idea

Mixer Edit Modes let you reorganise the X-Touch around a particular mixing task.

Instead of always seeing:

eight tracks

x

many different controls

you can temporarily see:

one kind of control

x

eight tracks

That change of perspective is what makes these modes useful.

The X-Touch is still an eight-channel control surface.

But DrivenByMoss lets those eight channels become eight simultaneous views of:

- volume;
- panorama;
- Sends;
- track parameters;

depending on what you need at that moment.

Once this way of thinking becomes natural, the controller begins to feel less like a fixed bank of knobs and faders and more like a surface that reorganises itself around the mix.

## Coming Next

Mixer Edit Modes help us change what we are controlling.

The next chapter deals with another question:

Where are we in the project?

Markers give us named positions in the timeline, and DrivenByMoss gives the X-Touch several ways to create, display and navigate them.

In the next chapter we will look at Markers and Advanced Navigation.

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## 19\_advanced\_recording\_and\_overdub

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## Chapter 20 - Towards a Mouse-Free (or Mouse-Lite) Workflow

If you have made it this far, you may have noticed something.

The mouse is spending rather more time sitting quietly on the desk.

That is no accident.

The purpose of the X-Touch is not necessarily to eliminate the mouse completely. Bitwig is a sophisticated graphical application, and there are many tasks for which a mouse remains an excellent tool.

But there is a considerable difference between using the mouse because it is the best tool for the job and using it simply because reaching for it has become a habit.

This chapter is about breaking that habit.

Not Mouse-Free at all costs.

Mouse-Lite.

## From Controls to Workflows

Throughout this book we have examined individual parts of the X-Touch.

We have looked at:

- banks and channels
- modes
- SELECT
- displays and feedback
- V-Pots
- motor faders
- transport controls
- Device Mode
- Browser Mode

Learning those features individually was necessary.

But nobody sits down to make music and thinks:

"I think I'll operate some V-Pots now."

You think:

"The bass is too loud."

Or:

"That synth needs more filter movement."

Or:

"I need a delay on this track."

The real goal is therefore not to learn the controls.

It is to combine them into workflows.

## The Controller Should Disappear

At first, using the X-Touch requires conscious thought.

You ask yourself:

- Which button do I press?
- Which mode do I need?
- What does this encoder do?
- Which bank contains that track?

That is completely normal.

With practice, those questions begin to disappear.

You hear that the bass is too loud and your hand moves to the appropriate fader.

You want to adjust an instrument and instinctively select the track and enter Device Mode.

You want to stop playback and your hand finds STOP without looking.

The controller has not become simpler.

You have stopped thinking about the controller.

That is the goal.

### Field Note

A well-learned control surface eventually becomes rather like a musical instrument.

You stop thinking about the mechanism and start thinking about the result.

## Workflow 1 — Balancing a Mix

Consider one of the most common tasks in any DAW:

Balancing track levels.

With a mouse, you might locate each channel in Bitwig's mixer and adjust its virtual fader individually.

With the X-Touch, the process becomes physical.

Start playback.

Listen.

Place your fingers on several faders.

Bring one instrument down.

Raise another slightly.

Perhaps reduce the vocal while increasing the drums.

You are no longer adjusting a sequence of individual graphical objects.

You are balancing several sounds together.

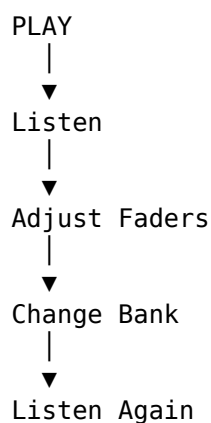
If the project contains more than eight tracks, change banks.

The scribble strips update.

The motor faders move to the levels of the newly visible tracks.

Continue mixing.

The workflow becomes:



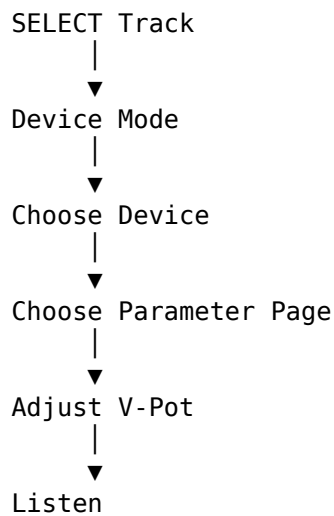
Notice how little attention this requires from the computer screen.

Your attention is primarily on the music.

## Workflow 2 — Adjusting a Device

Now imagine hearing a synthesizer whose filter needs adjustment.

The sequence is already familiar:



Earlier in the book, each of these steps was a separate concept.

Now they form one operation.

You are no longer thinking:

"I am using SELECT, followed by Device Mode, followed by a V-Pot."

You are thinking:

"I'm adjusting the synth."

That change in thinking is important.

## Workflow 3 — Adding an Effect

Suppose the vocal needs some delay.

Again, begin with focus.

Select the vocal track.

Open the Browser.

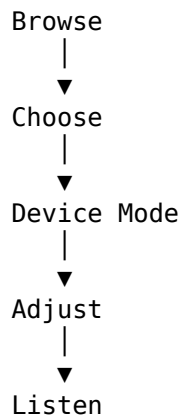
Locate an appropriate delay.

Choose it.

Move into Device Mode and adjust its parameters.

Conceptually:





What once required several trips between different areas of the Bitwig interface can increasingly be treated as a single workflow.

#### Field Note

The important measure of success is not how many mouse clicks you have eliminated.

It is how often you can remain focused on the musical task without interrupting yourself to operate the computer.

## Workflow 4 — Recording and Reviewing

Transport controls become particularly useful during recording.

Imagine recording a part.

You prepare the appropriate track and position the transport.

Then:

1. Press RECORD.
2. Perform the part.
3. Press STOP.
4. Navigate back to the beginning.
5. Press PLAY.
6. Listen.
7. Repeat if necessary.

This may sound almost trivial.

That is precisely the point.

The mechanics of operating the DAW begin to disappear.

You remain focused on performing and listening.



## Workflow 5 — Writing Automation

The motor faders provide another particularly tactile workflow.

Suppose a track needs to rise gradually during a section.

Rather than drawing the change with the mouse, you can perform it.

Prepare the appropriate automation state.

Start playback.

Touch the fader.

Move it as the music plays.

The gesture becomes automation.

When you play the project again, the motor fader can reproduce that movement.

This is an excellent example of the difference between editing music and performing a mix.

The result may ultimately be the same automation data.

The experience of creating it is very different.

## Keep Your Eyes Where They Matter

One unexpected benefit of a control surface is that it can change how you use the computer display.

Without the X-Touch, the screen must often perform two jobs:

1. Show you useful information.
2. Provide the controls needed to manipulate that information.

The X-Touch separates those roles.

The screen can show:

- the arrangement
- waveforms
- device interfaces
- automation
- meters
- other visual information

while your hands operate:

- faders
- V-Pots

- SELECT buttons
- transport controls
- navigation

The screen becomes more useful because you no longer need to keep pointing at everything you want to change.

## Listen More, Look Less

Music production software is extraordinarily visual.

That can sometimes encourage us to make decisions with our eyes.

A waveform looks too small.

A meter looks too low.

A knob appears to be in the wrong position.

But ultimately we are making sound.

Physical controls make it easier to perform some operations without staring at their graphical equivalents.

Move a fader while listening.

Turn an encoder while listening.

Balance two tracks while listening.

Choose a sound and then judge it by listening.

### Field Note

The X-Touch cannot tell you whether something sounds good.

Fortunately, you already own the equipment required for that job.

Your ears.

## When the Mouse Is Better

There is no prize for avoiding the mouse.

Some tasks are inherently graphical.

Detailed editing of audio events, arranging clips, drawing complex automation and interacting with visual plug-in interfaces may be quicker and clearer with a mouse or trackpad.

Use it.

A Mouse-Lite workflow is not a set of rules.

It is a change in instinct.

Instead of automatically asking:

"Where do I click?"

ask:

"What is the easiest way to do this?"

Sometimes the answer will be the X-Touch.

Sometimes it will be the mouse.

Sometimes it may be the computer keyboard.

The best workflow uses all of them appropriately.

#### Reality Check

If performing an operation from the X-Touch takes considerably longer than doing it with the mouse, use the mouse.

Mouse-Lite should make working with Bitwig easier, not turn mouse avoidance into a competitive sport.

## Build Muscle Memory Gradually

Do not attempt to stop using the mouse overnight.

That usually produces frustration rather than fluency.

Instead, choose a few operations that you perform frequently.

For example:

- PLAY and STOP
- track selection
- volume changes
- pan
- bank navigation

Use the X-Touch for those consistently.

Once they become automatic, add another workflow.

Perhaps:

- send adjustment
- Device Mode

- parameter control

Then add Browser Mode.

Over time, the number of occasions on which you instinctively reach for the mouse begins to shrink.

Not because you are deliberately avoiding it.

Because your hands already know a better way.

## When You Get Lost

Even experienced users occasionally find themselves wondering what the controller is doing.

When that happens, return to the mental model we have developed throughout this book.

Ask:

1. Which track is selected?
2. Which bank am I viewing?
3. Which mode am I in?
4. What are the scribble strips showing?
5. What do the LEDs tell me?
6. What does this control represent right now?

Do not start pressing buttons randomly.

The X-Touch is usually already telling you the answer.

Observe before you adjust.

That principle has appeared repeatedly throughout this book for a reason.

## The X-Touch Is Not Bitwig

There is one final distinction worth making.

The X-Touch does not replace Bitwig.

Nor should it.

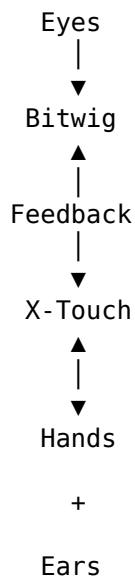
Bitwig remains the environment in which the project exists.

The X-Touch provides another way of interacting with that environment.

Some things are better represented visually.

Some things are better controlled physically.

The power comes from combining the two.



The computer provides the visual workspace.

The X-Touch provides the tactile workspace.

Your ears decide what matters.

## From Interface to Instrument

When we began, the X-Touch could easily have appeared to be an intimidating collection of:

- buttons
- displays
- knobs
- faders
- LEDs

We now know that this is not the most useful way to understand it.

The X-Touch is a context-sensitive surface.

Banks determine which channels are visible.

Modes determine what the controls mean.

SELECT determines focus.

The displays tell you what is happening.

V-Pots adapt to the current task.

Motor faders physically reflect Bitwig's state.

Transport controls allow the session to move beneath your hands.

Device Mode lets you reach inside the project.

Browser Mode lets you add to it.

None of these ideas is particularly complicated by itself.

Together, they transform the controller from a collection of controls into a coherent working environment.

## Towards Mouse-Free — or Mouse-Lite

You may eventually discover that entire stretches of a session take place without touching the mouse.

You start playback.

You select tracks.

You balance the mix.

You adjust sends.

You navigate devices.

You change parameters.

You stop.

You rewind.

You listen again.

And then, eventually, you need to perform a detailed graphical edit.

So you reach for the mouse.

That is not a failure to achieve a Mouse-Free workflow.

That is a successful Mouse-Lite workflow.

The goal was never to remove a useful tool from your desk.

The goal was to add another one.

## Where You Go From Here

There is much more that can be explored.

DrivenByMoss provides capabilities beyond the foundations covered in this book, and both Bitwig and the X-Touch reward experimentation.

But you now have something more useful than a memorised list of commands.

You have a mental model.

When you encounter an unfamiliar feature, ask:

- What is selected?
- What is the current mode?
- What are the controls representing?
- What feedback is the X-Touch providing?

Once you can answer those questions, unfamiliar operations become much easier to understand.

The aim of this book has never been to teach you every button combination.

It has been to help you reach the point where you no longer need someone to explain every button combination.

## Exercise

For the final exercise, open a Bitwig project you know well.

Spend fifteen or twenty minutes working normally, but pay attention whenever you reach for the mouse.

Do not forbid yourself from using it.

Instead, pause briefly and ask:

> "Could I do this more naturally from the X-Touch?"

If the answer is yes, try it.

If the answer is no, use the mouse and continue.

During the session, try to perform as many of these tasks as practical from the controller:

1. Navigate the project.
2. Start and stop playback.
3. Select tracks.
4. Change banks.
5. Balance track levels.
6. Adjust pan or sends.
7. Navigate to a device.
8. Adjust device parameters.
9. Browse for something new.
10. Return to the mixer and continue working.

Do not count mouse clicks.

Do not measure how long you remain "mouse-free".

Listen to the music and notice whether operating the computer is interrupting your concentration less often.

That is the real test.

The best control surface is not the one with the most features.

It is the one that lets you forget that you are operating a control surface.



## One Last Thought

The X-Touch is covered with controls.

Learning it can therefore appear to be an exercise in remembering what all those controls do.

It isn't.

The real skill is learning to understand context, focus and feedback.

Once those ideas become familiar, the buttons stop looking like a collection of things to memorise.

They become tools.

And when the tools become familiar enough, something even better happens.

You stop thinking about them.

You simply make music.

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